Broadband Transient Localization over WDM Fibre: Per-Pair Limits and Empirical Fusion

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Abstract

We study range estimation from the wavelength-dependent polarization signature of a short transient in a wavelength-division-multiplexed (WDM) fibre. Chromatic dispersion maps range to relative channel delay. We derive a two-sensor effective-SNR Cramér–Rao bound (CRB) for each pair, then evaluate channel selection, dispersion weighting, robust fusion, and delay-estimator bias in Monte Carlo simulations. On an equispaced grid, all-pair D^2 weighting is close to hard-filtered equal averaging, while robust fusion can give further gains. An alternating band-edge grid behaves differently because its low-dispersion pairs receive almost no D^2 weight. Pair estimates share channels and are correlated, so fused performance is reported empirically rather than as a combined CRB.

1 Problem and contribution

Distributed fibre sensing can detect mechanical disturbances through changes in the received state of polarization (SOP) [1]. When a transient has a measurable, temporally identifiable signature in several WDM channels, chromatic dispersion (CD) makes the signatures arrive at different times. For channels i and j, let d_i and d_j be their CD delay per unit distance; then the signed relative-delay slope is $\Delta d_{ij} = d_i - d_j$, the pair delay at range Z is $\tau_{ij} = Z\Delta d_{ij}$, and a measured delay gives $\hat{Z}_{ij} = \hat{\tau}_{ij}/\Delta d_{ij}$. Thus wavelength diversity supplies multiple range-sensitive pairs, but not independent measurements: pairs that share a channel also share its noise and waveform-specific errors.

This study addresses three distinct questions: the timing limit for one pair, how channel placement changes that pair's distance sensitivity, and how to combine the correlated pair estimates. The CRB answers the first; Monte Carlo RMSE, bias, and spread assess fusion. We do not claim that summing pairwise CRBs gives a multichannel bound. The method is scoped to events with a detectable common cross-channel signature; it is not validated for arbitrary event types.

2 Model and per-pair timing limit

The simulated SOP is represented by two real Stokes components. A transient modulates a local PMD section of length 0.5 km, centered 500 km from the receiver; the remaining path contributes deterministic wavelength-dependent CD delays, not a distributed PMD cascade. This isolates the local perturbation response from long-path CD. The observation model follows standard delay-estimation and PMD treatments [2, 3, 4].

For two noisy records, the frequency-bin information depends on both channels' SNRs. At frequency f, their effective SNR is $\mathrm{SNR}_{\mathrm{eff}}(f) = \rho_i(f)\rho_j(f)/[1+\rho_i(f)+\rho_j(f)]$. For equal channels it is $\rho^2/(1+2\rho)$: approximately ρ^2 at low SNR, when both observations must contain signal, and $\rho/2$ at high SNR, when neither channel is a noiseless timing reference. Each bin contributes in proportion to $(2\pi f)^2\mathrm{SNR}_{\mathrm{eff}}$; the contributions from both independent SOP components and all frequency bins add to the pair information J_{ij}. Thus $\mathrm{Var}(\hat{\tau}_{ij}) \geq 1/J_{ij}$, and the corresponding range standard deviation is $1/(|\Delta d_{ij}|\sqrt{J_{ij}})$. The frequency weighting makes broadband temporal structure especially useful for timing [2, 3].

The clean simulated records supply the signal spectra; the selected noise variance sets the per-bin noise powers. The CRB is therefore a model-based reference, not a fit to a measured delay error or cross-correlation peak. In a controlled two-component Gaussian-pulse check, simulated pair-delay spread was within about 4% of the CRB at 60 dB energy SNR. In a representative production cell, the median pair spread was 1.43 times the per-pair CRB standard deviation (10th–90th percentiles: 1.08–1.87), reflecting waveform variation and estimator effects.

3 Pair selection and fusion

For a set of pair range estimates, equal fusion uses $w_{ij} = 1$. Since a fixed timing error maps to a smaller range error for larger $|\Delta d_{ij}|$, dispersion weighting uses $w_{ij} = D_{ij}^2$, where D_{ij} is the integrated CD magnitude in ps/km. This is an inverse-marginal-variance heuristic under comparable pair timing errors, not a covariance-aware optimum. A hard filter retains pairs satisfying $D_{ij}^2 \geq \alpha \max D^2$; the default $\alpha = 0.25$ is equivalent to retaining pairs with $|D_{ij}| \geq 0.5 D_{\max}$.

Robust fusion multiplies the geometry weight by a Tukey-bisquare residual weight: $w_{ij} = D_{ij}^2\psi(r_{ij})$, where r_{ij} is the distance from the pair estimate to the pool median, $s = 1.5\,\mathrm{IQR}$, and $\psi(r) = [1-(r/(2.5s))^2]^2$ for $r < 2.5s$ and zero otherwise. Thus D^2 favors pairs that are more distance-sensitive, while the residual factor suppresses pair estimates that disagree with the observed consensus. Robust fusion can help when most pairs form a reliable consensus, but may fail if that consensus is biased or wrong.

4 Simulation results

The production study used 10 independent PMD realizations and 100 noise realizations per fibre, two SOP components, and channel subsets from a 241-channel, 50-GHz master grid. Nominal envelope bandwidth and per-component

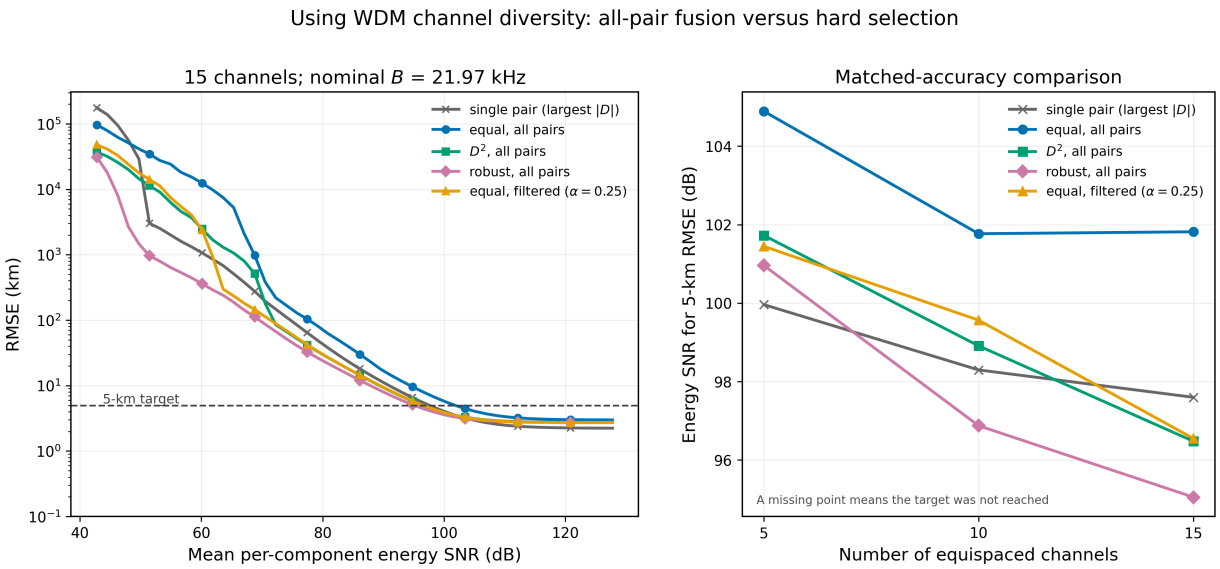


Figure 1: EQUISPACED GRID: empirical RMSE and energy SNR needed for a 5-km RMSE target as channel count increases. The best method depends on the operating point; pair estimates are correlated.

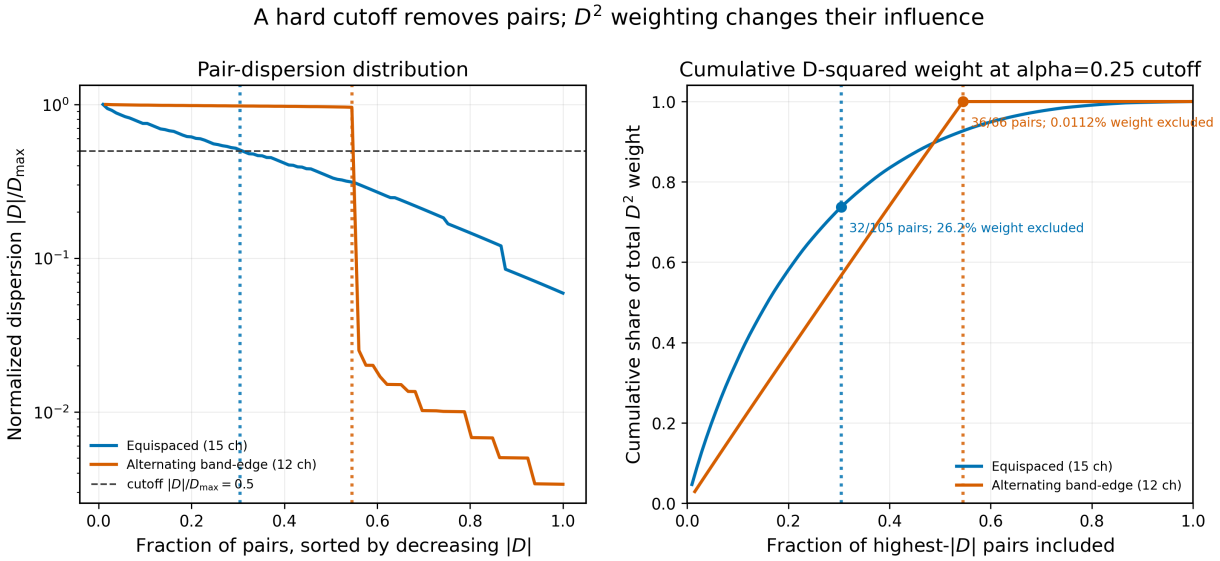


Figure 2: Pair-dispersion distributions and cumulative D^2 weight. A graded equispaced grid retains appreciable weight below the hard cutoff; the band-edge layout nearly separates useful and weak pairs.

energy SNR were swept; the main fusion comparisons use 5, 10, and 15 equispaced channels, with an alternating band-edge subset as contrast. RMSE includes both estimator bias and random spread. The quoted SNR crossings are interpolated point estimates; paired uncertainty intervals have not yet been calculated.

Figure 1 summarizes the equispaced-grid comparison. For 15 channels near $B_{\mathrm{nom}} = 22\,\mathrm{kHz}$, the per-component energy SNR required for 5-km RMSE was 101.82 dB for equal averaging over all pairs, 96.48 dB for all-pair D^2 weighting, 95.05 dB for robust all-pair fusion, and 96.55 dB for equal averaging after the $\alpha = 0.25$ filter. At this operating point, soft D^2 weighting is close to hard selection; robust fusion improves the crossing by about 1.5 dB relative to filtered equal averaging. These are empirical point comparisons, not evidence that every small difference is statistically resolved.

The mechanism is visible in Fig. 2. On the equispaced grid, pair dispersion varies gradually: the hard filter retains 32 of 105 pairs, while the excluded pairs still account for 26.2% of the all-pair D^2 weight. On the 12-channel alternating band-edge grid, dispersion separates into low- and high-sensitivity groups; the filter retains 36 of 66 pairs, and excluded pairs have only 0.011% of total D^2 weight. Consistent with this, at about 22 kHz the band-edge 12-channel 5-km target required 119.84 dB for equal all-pair fusion but about 90.0 dB for D^2, robust, or filtered-equal fusion. Here D^2 weighting nearly acts as a soft version of deleting the low-dispersion group. The weight fractions describe the rule's allocation, not independent statistical information.

5 Estimator choice, implications, and limits

Fusion cannot repair every pair-delay estimator failure. Figure 3 compares parabolic peak interpolation with generalized cross-correlation with phase transform (GCC-PHAT) [5], both under robust fusion. Their ranking changes

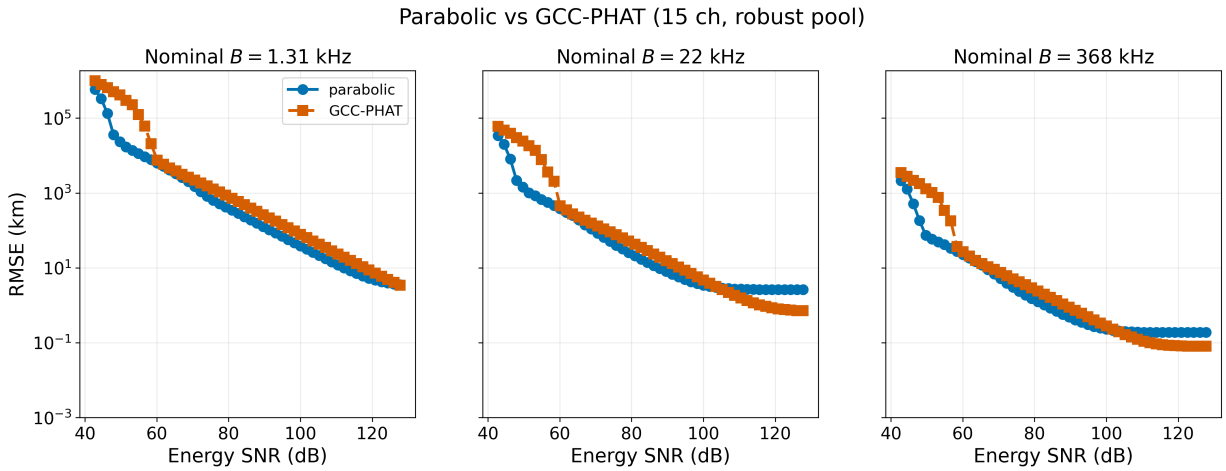


Figure 3: Parabolic and GCC-PHAT RMSE after the same robust fusion on an equispaced 15-channel grid. The crossover depends on bandwidth and SNR; neither estimator dominates throughout.

with bandwidth and SNR. Near 22 kHz and 127.8 dB, the 15-channel RMSE was 2.64 km for parabolic and 0.71 km for GCC-PHAT; near 1.3 kHz and 103.8 dB, parabolic was better (25.8 versus 52.3 km). Parabolic interpolation often has lower variance but can retain a bias floor; GCC-PHAT can reduce that bias but is not universally unbiased or robust to weak spectral bins. Accordingly, RMSE—not variance alone—is reported.

The tested high-SNR spread approximately follows $\sigma_Z \propto (\sqrt{\mathrm{SNR}_E} B_{\mathrm{nom}})^{-1}$; more channels often reduce the SNR required for a target error, while averaging independent fibre realizations reduces spread approximately as $N_{\mathrm{fib}}^{-1/2}$. These are empirical scaling trends, not universal laws. The results support a practical sequence: confirm cross-channel observability, inspect the pair-dispersion distribution, compare equal, D^2, and robust fusion, and select the delay estimator using both bias and spread. Pair overlap precludes treating the $N(N-1)/2$ estimates as independent.

This is a simulation study of one localized SOP-perturbation model, with additive white independent noise and no experimental validation. Distributed PMD outside the local section, receiver imperfections, dispersion-calibration uncertainty, and paired confidence intervals remain future work. The main conclusion is conditional: WDM diversity can improve localization, but the gain depends on wavelength placement, estimator behavior, and fusion; no universal combined-CRB attainment is claimed.

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